

DSA0601 DATA HANDLING AND VISUALIZATION

Activity-3

Question-1

Activity 1: Heatmap – Student Performance Analysis

Student	Mathematics	Physics	Chemistry	Programming	English
S1	92	88	84	95	81
S2	76	72	69	80	74
S3	81	78	83	85	79
S4	65	70	68	72	75
S5	90	94	91	96	89
S6	58	62	60	65	70
S7	84	86	82	88	80
S8	71	75	73	77	72
S9	95	91	93	98	94
S10	79	81	77	83	78

Questions:

1. Construct a Heatmap.
2. Identify highest overall performer.
3. Find highest average subject.
4. Identify students needing support.
5. Explain Heatmap advantages.

R Code

```
install.packages("pheatmap")  
library(pheatmap)  
students <- data.frame(  
  Mathematics=c(92,76,81,65,90,58,84,71,95,79),  
  Physics=c(88,72,78,70,94,62,86,75,91,81),  
  Chemistry=c(84,69,83,68,91,60,82,73,93,77),  
  Programming=c(95,80,85,72,96,65,88,77,98,83),  
  English=c(81,74,79,75,89,70,80,72,94,78)  
)
```

```

rownames(students) <- paste0("S",1:10)

pheatmap(students,
         color=colorRampPalette(c("yellow","orange","red"))(100),
         main="Student Performance Heatmap")

avg_student <- rowMeans(students)
cat("Highest Overall Performer:", names(which.max(avg_student)), "\n")

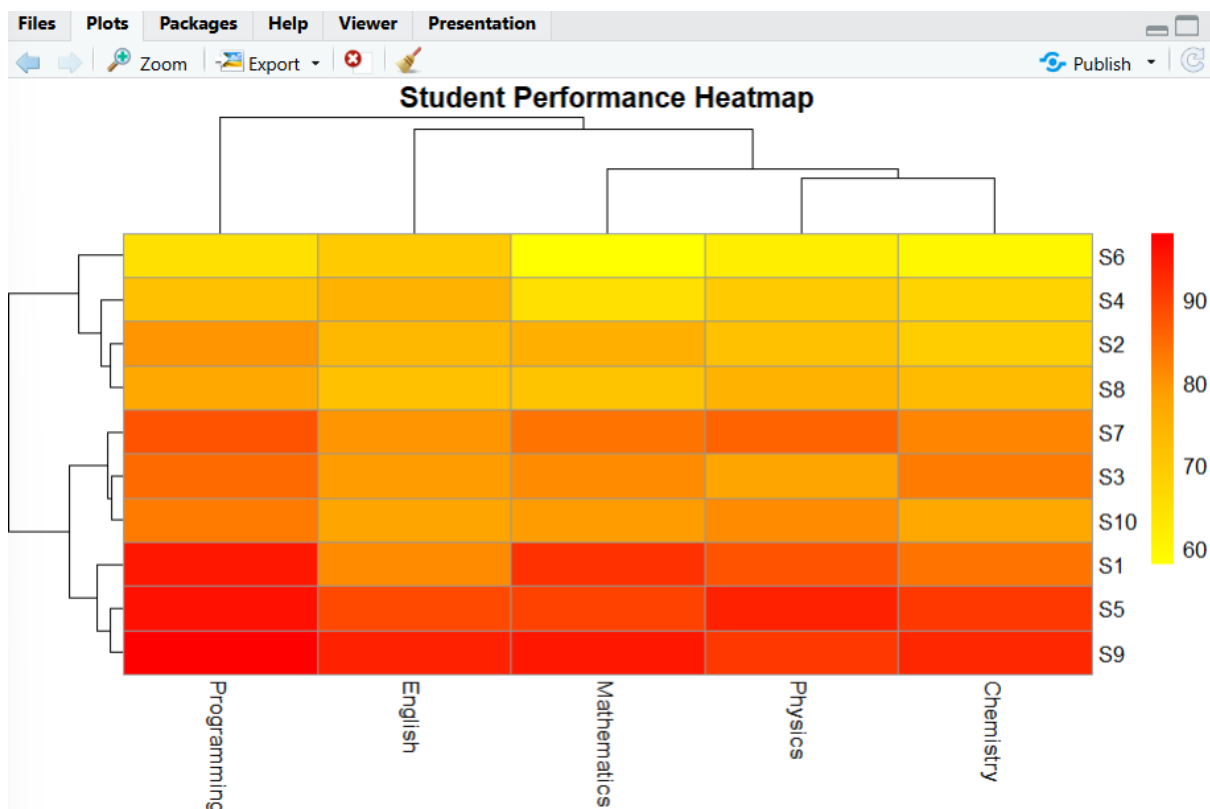
avg_subject <- colMeans(students)
cat("Highest Average Subject:", names(which.max(avg_subject)), "\n")

support <- names(avg_student[avg_student < 70])
cat("Students Needing Support:", support, "\n")

print(avg_subject)

```

1.



2. Highest Overall Performer: S9

3. Highest Average Subject: Programming

4. Students Needing Support: S6

5. Heatmap Advantages:

- Displays large datasets using colour intensity.
- Makes comparison between students and subjects easy.
- Quickly identifies high and low scores.
- Highlights patterns, trends, and outliers.
- Easy to understand and visually appealing.

Question-2

Activity 2: Violin Plot & Boxplot – Employee Salary Analysis

Employee	HR	IT	Finance	Marketing
E1	4.2	6.5	5.8	5.1
E2	4.5	7.0	6.2	5.3
E3	4.7	7.4	6.0	5.5
E4	4.1	6.8	5.9	5.4
E5	4.6	8.2	6.7	5.8
E6	4.3	7.6	6.4	5.6
E7	4.8	8.5	6.9	5.9
E8	4.4	7.1	6.1	5.2
E9	4.9	8.8	7.2	6.0
E10	4.5	7.3	6.3	5.4

Questions:

1. Draw violin plots.
2. Construct boxplots.
3. Largest variation?
4. Highest median?
5. Which plot better represents distribution?

R CODE

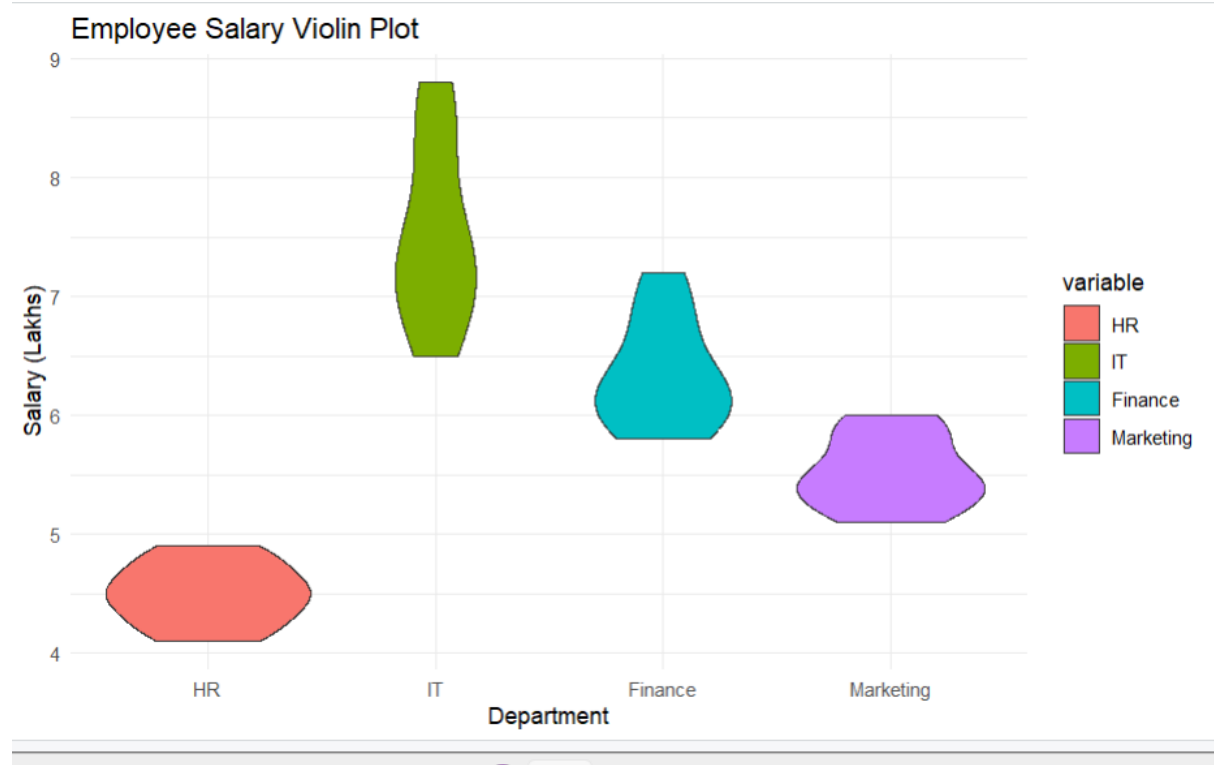
```
install.packages("reshape2")  
  
library(reshape2)  
  
salary <- data.frame(  
  HR=c(4.2,4.5,4.7,4.1,4.6,4.3,4.8,4.4,4.9,4.5),  
  IT=c(6.5,7.0,7.4,6.8,8.2,7.6,8.5,7.1,8.8,7.3),
```

```

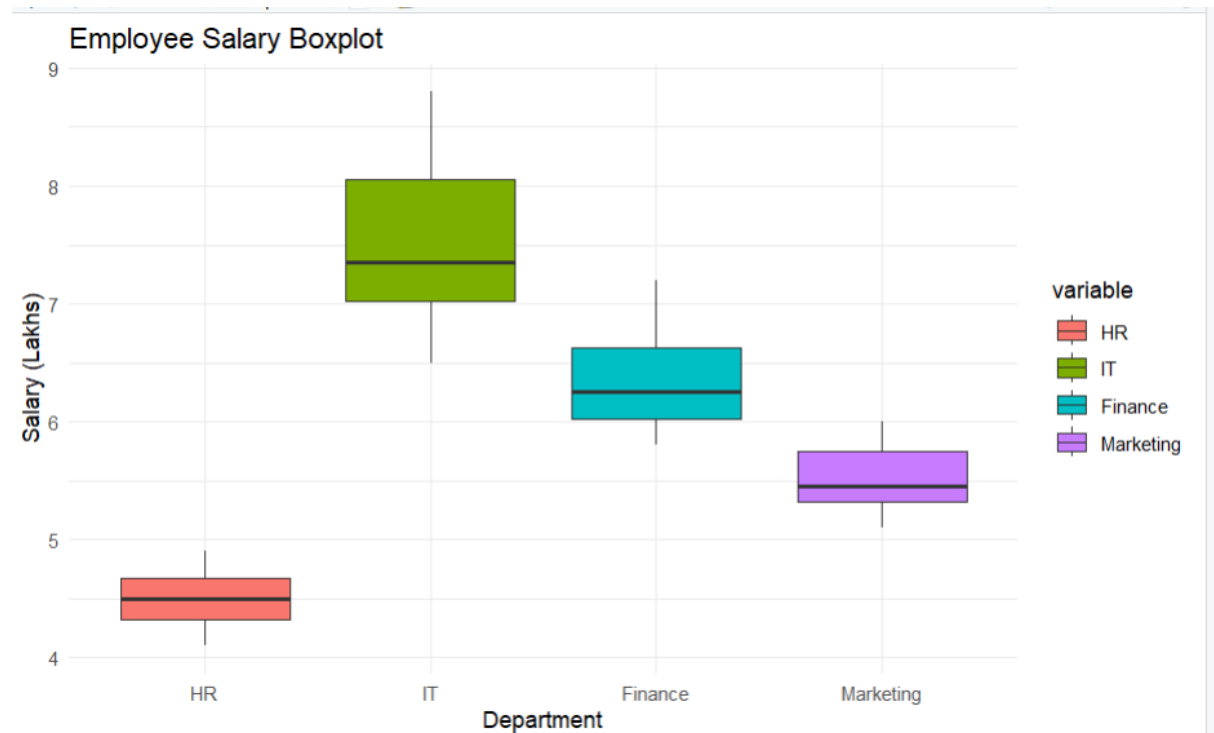
Finance=c(5.8,6.2,6.0,5.9,6.7,6.4,6.9,6.1,7.2,6.3),
Marketing=c(5.1,5.3,5.5,5.4,5.8,5.6,5.9,5.2,6.0,5.4)
)
salary_long <- melt(salary)
install.packages("ggplot2")
library(ggplot2)
ggplot(salary_long,
       aes(x=variable,y=value,fill=variable))+
  geom_violin()+
  theme_minimal()+
  labs(title="Employee Salary Violin Plot",
       x="Department",y="Salary (Lakhs)")
ggplot(salary_long,
       aes(x=variable,y=value,fill=variable))+
  geom_boxplot()+
  theme_minimal()+
  labs(title="Employee Salary Boxplot",
       x="Department",y="Salary (Lakhs)")
variation <- sapply(salary,sd)
cat("Largest Variation:",names(which.max(variation)),"\n")
median_salary <- sapply(salary,median)
cat("Highest Median:",names(which.max(median_salary)),"\n")
print(variation)
print(median_salary)

```

1.



2.



3. Largest Variation: IT

4. Highest Median: IT

5. Which plot better represents the distribution?

Answer: Violin Plot

Explanation:

- Shows the complete distribution shape.
- Displays data density at different values.
- Helps identify multiple peaks (multimodal data).
- More informative than a boxplot for understanding distribution.
- Suitable for comparing distributions across groups.

Question-3

Activity 3: Histogram & Density Plot – Website Response Time

Obs	Response Time (ms)
1	210
2	225
3	230
4	240
5	250
6	260
7	270
8	280
9	285
10	290
11	300
12	305
13	310
14	315
15	320
16	330
17	340
18	350
19	360
20	380

Questions:

1. Draw histogram.
2. Plot density.
3. Determine skewness.
4. Common range.
5. Which plot better represents distribution?

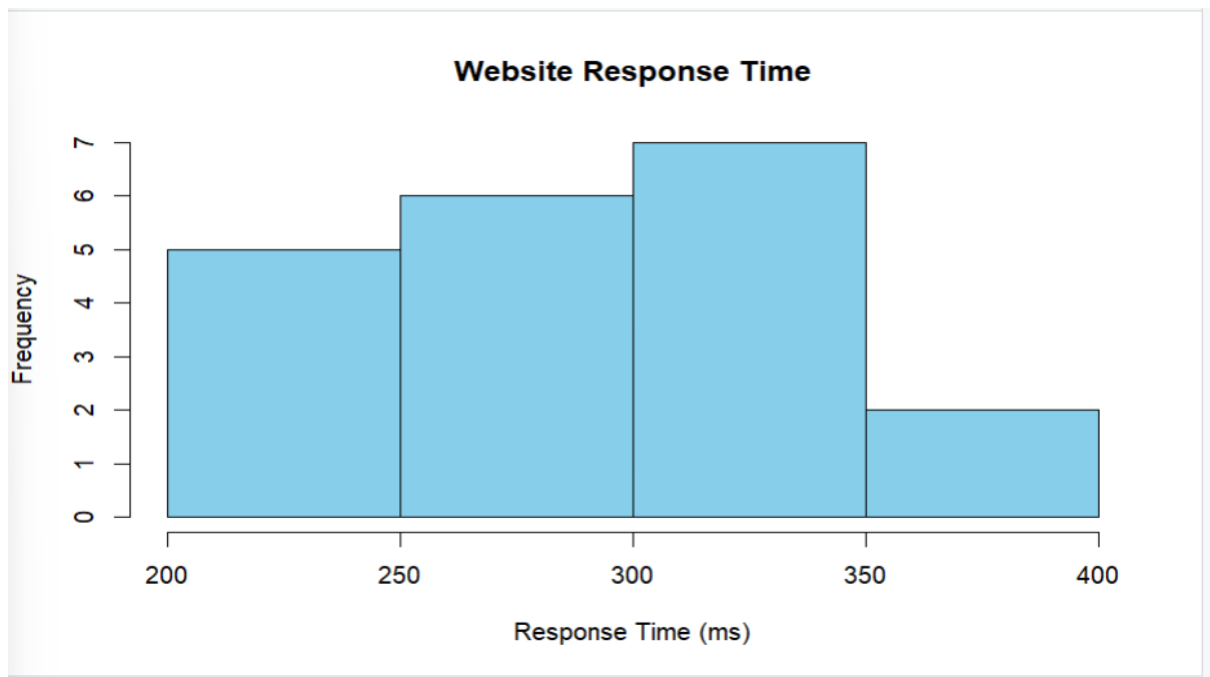
R CODE

```
response <- c(210,225,230,240,250,260,270,280,285,290,
```

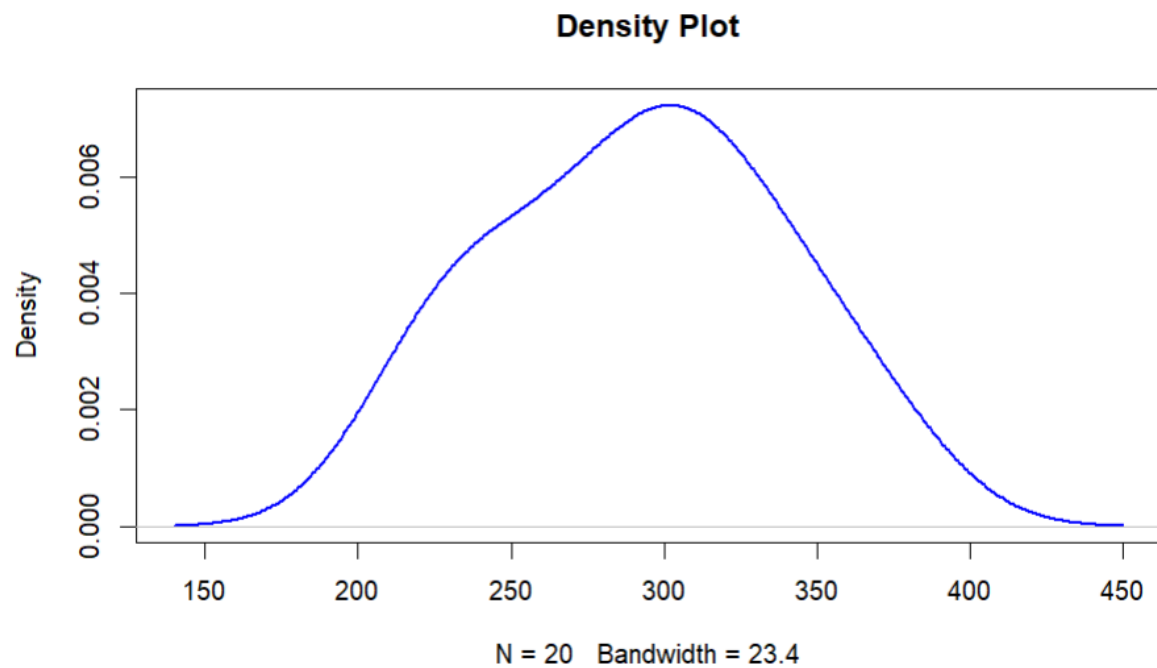
```
300,305,310,315,320,330,340,350,360,380)
```

```
hist(response,  
      col="skyblue",  
      main="Website Response Time",  
      xlab="Response Time (ms)")  
plot(density(response),  
      main="Density Plot",  
      col="blue",  
      lwd=2)  
install.packages("moments")  
library(moments)  
cat("Skewness =",skewness(response),"\n")  
cat("Most response times are between 250 and 330 ms.\n")  
summary(response)
```

1.



2.



3. Skewness = -0.007827051

4. Most response times are between 250 and 330 ms.

5. Which plot better represents the distribution?

Answer: Density Plot

Points:

- Provides a smooth representation of data distribution.
- Removes the effect of histogram bin sizes.
- Makes identifying skewness easier.
- Better for comparing multiple datasets.
- Clearly shows the overall trend of the data.

Question-4

Activity 4: Ridgeline Plot – Electricity Consumption

House	Jan	Feb	Mar	Apr	May
H1	220	235	250	275	295
H2	240	245	260	285	305
H3	230	238	255	280	300
H4	215	228	248	270	292
H5	225	240	258	282	298
H6	245	250	265	290	310
H7	235	242	259	284	302
H8	228	236	252	278	297
H9	238	246	262	287	307
H10	232	239	256	281	301

Questions:

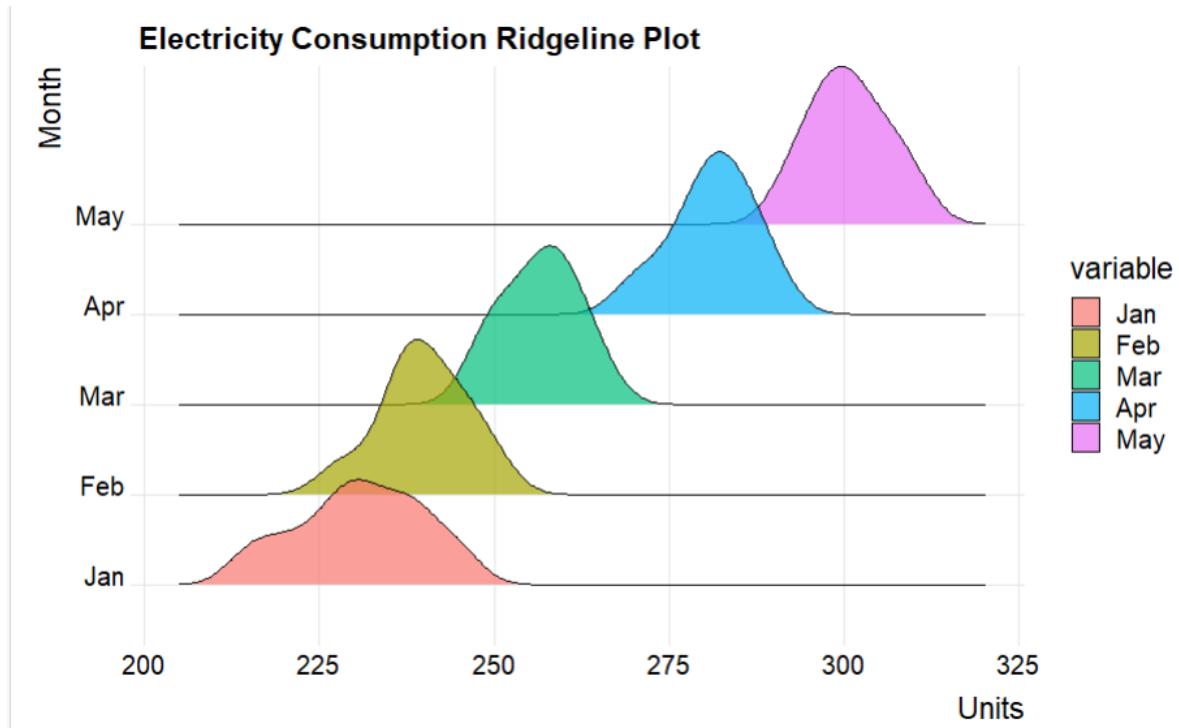
1. Create ridgeline plot.
2. Highest average month?
3. Widest spread?
4. Describe trend.
5. Why ridgeline?

RCODE

```
install.packages("ggridges")
install.packages("reshape2")
library(ggridges)
library(reshape2)
library(ggplot2)
electricity <- data.frame(
  Jan=c(220,240,230,215,225,245,235,228,238,232),
  Feb=c(235,245,238,228,240,250,242,236,246,239),
  Mar=c(250,260,255,248,258,265,259,252,262,256),
  Apr=c(275,285,280,270,282,290,284,278,287,281),
  May=c(295,305,300,292,298,310,302,297,307,301)
)
```

```
electricity_long <- melt(electricity)
ggplot(electricity_long,
       aes(x=value,y=variable,fill=variable))+
geom_density_ridges(alpha=0.7)+
theme_ridges()+
labs(title="Electricity Consumption Ridgeline Plot",
      x="Units",y="Month")
avg_month <- colMeans(electricity)
cat("Highest Average Month:",
    names(which.max(avg_month)),"\n")
spread <- sapply(electricity,sd)
cat("Widest Spread:",
    names(which.max(spread)),"\n")
print(avg_month)
print(spread)
```

1.



2. Highest Average Month: May

3. Widest Spread: Jan

4. Describe the trend

- Electricity consumption increases from January to May.
- Every month shows high average usage than the previous month.
- No sudden decrease is observed.
- Indicates a consistent upward trend.
- May has the highest overall consumption.

5. Why use a Ridgeline Plot?

- Compares distributions across multiple months.
- Saves space while displaying many distributions.
- Makes trends and changes easy to identify.
- Clearly shows overlapping distributions.
- Provides an attractive and informative visualisation.

Question-5

Activity 5: Histogram, Density & Boxplot – Daily Steps

Person	20–30	31–45	46–60
P1	9500	8200	6500
P2	10200	8600	6700
P3	9800	8400	6900
P4	11000	9000	7200
P5	10500	8900	7000
P6	9700	8500	6800
P7	10100	8700	7100
P8	10800	9100	7300
P9	11200	9200	7400
P10	9900	8600	6950

Questions:

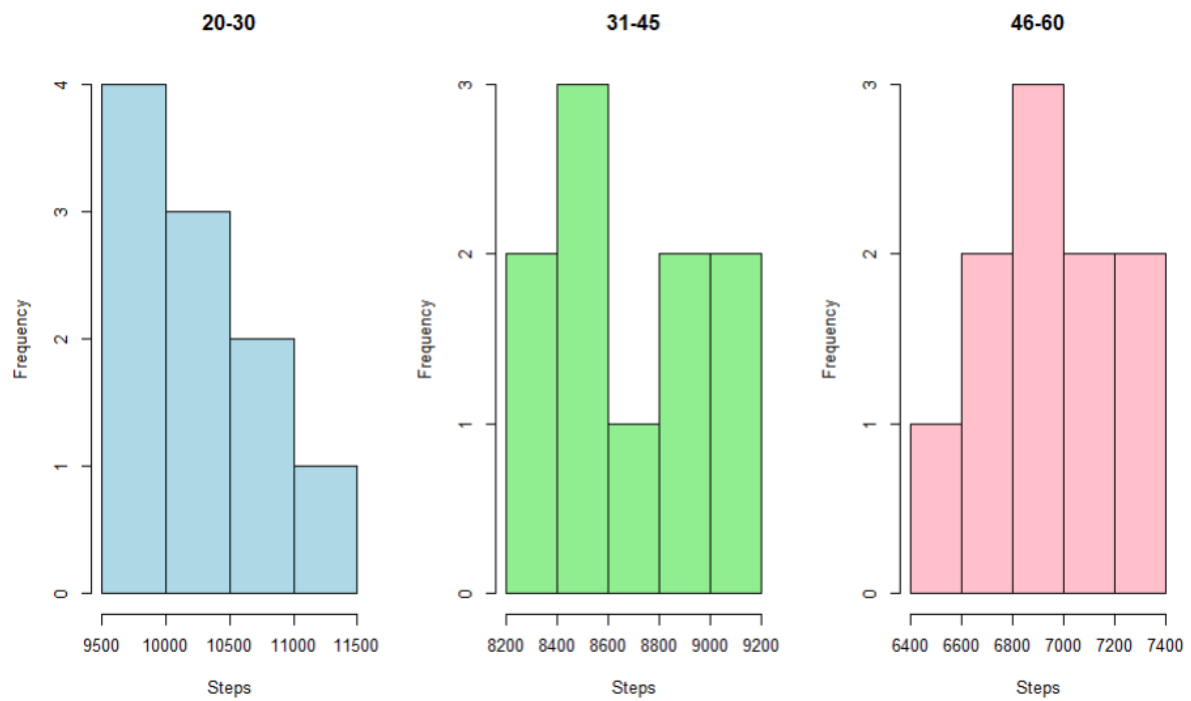
1. Draw histograms.
2. Plot density curves.
3. Create boxplots.
4. Highest median group?
5. Best visualization and justify.

R CODE

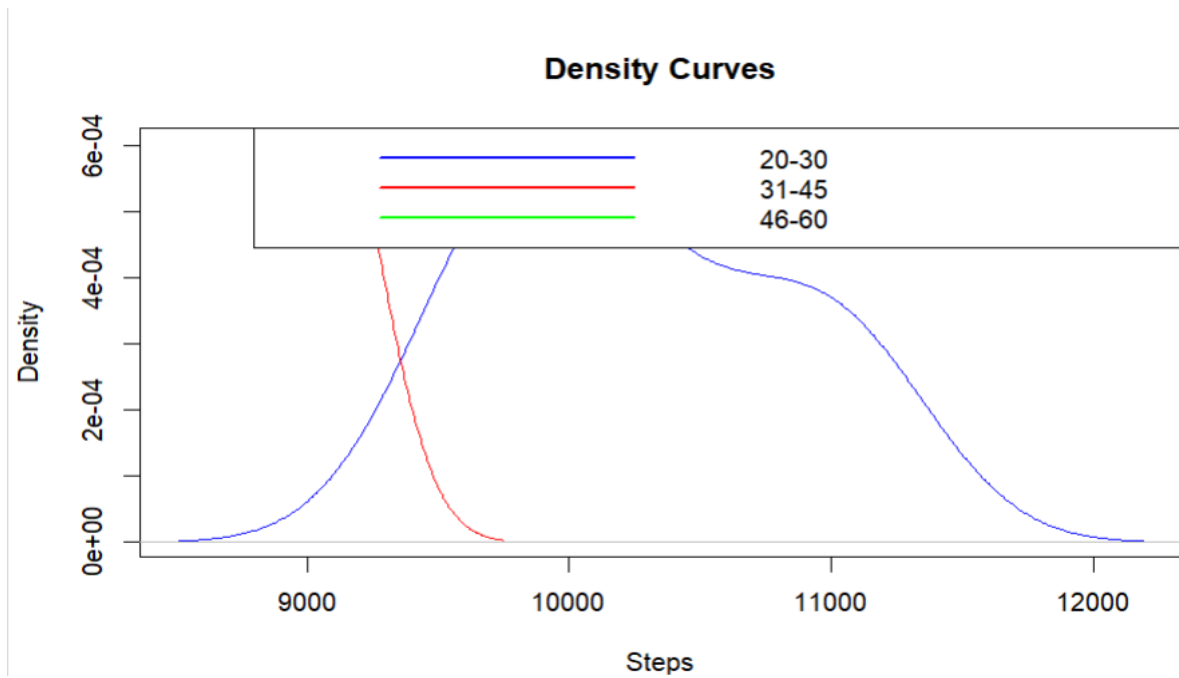
```
steps <- data.frame(  
  Age20_30=c(9500,10200,9800,11000,10500,9700,10100,10800,11200,9900),  
  Age31_45=c(8200,8600,8400,9000,8900,8500,8700,9100,9200,8600),  
  Age46_60=c(6500,6700,6900,7200,7000,6800,7100,7300,7400,6950)  
)  
  
par(mfrow=c(1,3))  
hist(steps$Age20_30,col="lightblue",  
  main="20-30",xlab="Steps")  
hist(steps$Age31_45,col="lightgreen",  
  main="31-45",xlab="Steps")  
hist(steps$Age46_60,col="pink",  
  main="46-60",xlab="Steps")  
  
par(mfrow=c(1,1))  
plot(density(steps$Age20_30),
```

```
col="blue",
main="Density Curves",
xlab="Steps")
lines(density(steps$Age31_45),col="red")
lines(density(steps$Age46_60),col="green")
legend("topright",
      legend=c("20-30","31-45","46-60"),
      col=c("blue","red","green"),
      lwd=2)
boxplot(steps,
      col=c("skyblue","lightgreen","pink"),
      main="Daily Steps Comparison")
median_steps <- sapply(steps,median)
cat("Highest Median Group:",
    names(which.max(median_steps)),"\n")
print(median_steps)
```

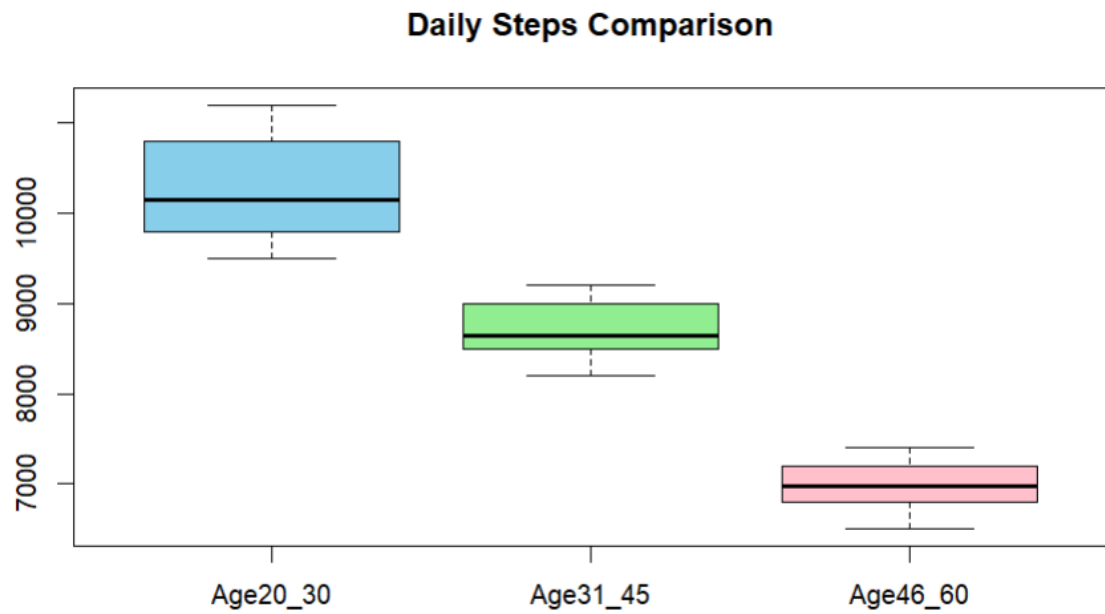
1.



2.



3.



4. Highest Median Group: Age20_30

5. Best visualization and justify

Answer: Boxplot

Points:

- Clearly compares medians of all groups.
- Displays spread (variability) of the data.
- Detects outliers easily.
- Simple and easy to interpret.
- Best for comparing multiple groups in one graph.